

Post-Implementation Report

RG Sonic32 v1-mini

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Toolchain	AMD Vivado ML Edition 2025.1
Result	Pass. Implementation complete with no errors.

*This document is a cumulative summary of the reports provided by Vivado
post-implementation*

1. Overview

This document presents the post-implementation analysis of the RG Sonic32 CPU v1-mini, including reports generated by the Vivado toolchain. The v1-mini configuration represents a reduced version of the full v1 core, with non-essential I/O removed to simplify initial hardware validation and performance characterization.

In its current form, v1-mini exposes only three external signals: clk, rst_n, and halt_out. Notably, it omits the UART interface present in the full v1 design, which is intended for program loading and register state inspection.

The complete v1 core will extend this architecture by integrating a UART module capable of reading hexadecimal program data from a defined source and outputting internal register states. This addition is intended to streamline debugging, verification, and system-level testing. Any further architectural or implementation differences between v1 and v1-mini will be documented in the corresponding post-implementation report for the full v1 system.

This report analyzes key implementation metrics, including design rule checks (DRC), methodology compliance, power consumption, timing performance, and resource utilization.

2. Design Configuration

Top-Level Module	cpu_top.v
Target Hardware	xc7a35tftg256-2 (Artix-7)
Design State	Fully Routed
Clock	clk: 14.286 ns (~70 MHz)
Ports	3 (clk IN, rst_n IN, halt_out OUT)

3. Timing Results

All timing constraints were successfully met for this implementation, with positive slack observed across all analyzed paths. No setup, hold, or pulse width violations were detected, indicating a timing-clean design at the target operating frequency.

CHECK	SLACK	FAILING ENDPOINTS	RESULT
Setup (WNS)	+0.260 ns	0 / 3000	PASS
Hold (WHS)	+0.140 ns	0 / 3000	PASS
Pulse Width (WPWS)	+6.643 ns	0 / 1282	PASS

Critical Paths

The worst setup path runs through the ALU multiply chain and terminates at the flags register.

Source: u_control_unit/ir_reg_reg[30] (FDRE, clocked by clk)
Destination: u_datapath/flags_reg_reg[0] (FDRE, clocked by clk)
Data Delay: 14.047 ns (logic 7.451 ns 53% | route 6.596 ns 47%)
Logic Levels: 17 (CARRY4=4, DSP48E1=2, LUT2=1, LUT4=1, LUT5=2, LUT6=5, MUXF7=1, MUXF8=1)
Slack: +0.260 ns

4. Resource Utilization

RESOURCE	USED	AVAILABLE	UTIL (%)
Slice LUTs	1605	20800	7.72%
Slice Registers	1249	41600	3.00%
Block RAM Tiles	16	50	32.00%
DSP48E1	3	90	3.33%
Bonded IOBs	3	170	1.76%
Slices	685	8150	8.40%
BUFGCTRL	1	32	3.13%

Resource usage is dominated by BRAM (~32%), with minimal utilization of LUTs and DSPs. This reflects a memory-heavy design with relatively low computational density in the current v1-mini configuration.

5. Power Analysis

The design consumes approximately 95 mW of power, dominated primarily by static power from the FPGA fabric, which accounts for 73% of total power.

COMPONENT	POWER (W)	NOTES
Block RAM	0.022	Dominant dynamic consumer
Clocks	0.003	Single global clock (BUFG)
Slice Logic	< 0.001	LUTs, FFs, carry chains
Signals	< 0.001	Routing net power
DSPs	0.000	ALU multiply units
I/O	< 0.001	clk, rst_n, halt_out
Device Static	0.069	Fixed FPGA leakage
Total On-Chip	0.095	Dynamic 0.026 W + Static 0.069 W

Given the low total power (~95 mW), thermal impact is negligible and does not impose any design constraints.

Power confidence is medium and estimates are based on default activity assumptions in Vivado and should be considered approximate pending post-silicon measurement or activity-based simulation.

6.Design Rule Check (DRC) Results

The Vivado DRC returned 0 errors and 0 critical warnings. All 12 checks are informational warnings.

RULE	COUNT	DESCRIPTION
DPIP-1	6	DSP input not pipelined
DPOP-1	3	DSP PREG output not pipelined
DPOP-2	3	DSP MREG output not pipelined

All warnings relate to the DSP48E1 blocks in the ALU multiplier path. Since pipelining is planned for v2, these warnings are deferred for version 2.

7.Methodology Check Report

37 methodology checks were reported, all advisory. No errors were found. The warnings fall into 4 categories:

RULE	COUNT	DESCRIPTION
SYNTH-6	16	BRAM output register not merged, read timing may be sub-optimal
SYNTH-10	3	Wide multiplier decomposed into multiple DSP blocks
SYNTH-15	16	Byte-wide write enable not inferred (address width > threshold)
TIMING-18	2	Missing input/output delay on rst_n and halt_out

The 16 SYNTH-6 instances each correspond to one of the 16 RAMB36E1 tiles in the memory subsystem. The root cause is that no output register was merged into the BRAM block during synthesis. Restructuring the memory read path to allow output register merging (by registering the read data inside the BRAM primitive rather than outside it) would improve read latency and timing margin. This is a structural HDL change and is deferred to v2.

8. Open Issues / Future Work

#	SOURCE	DESCRIPTION	PRIORITY
OI-01	DRC / Methodology	DSP48 inputs and outputs are not pipelined. Add AREG/BREG/MREG/PREG pipeline stages to improve timing headroom and power.	Medium
OI-02	Methodology	BRAM output registers not merged into block (SYNTH-6 x16). Restructure memory read path to allow register merging for better read timing.	Medium
OI-03	Methodology	Byte-wide write enable (BWWE) not inferred for BRAM (SYNTH-15 x16). Address width exceeds threshold. Consider ram_decomp=power attribute if BWWE is required.	Low
OI-04	Timing / Methodology	Missing set_input_delay on rst_n and halt_out (TIMING-18). Add I/O delay constraints to XDC for a fully closed design.	Low
OI-05	Power	Power confidence level is medium. Provide a simulation activity file (SAIF) or annotate switching activity in Vivado for a high-confidence power number.	Low

9. Conclusion

The post-implementation results for the RG Sonic32 CPU v1-mini demonstrate that the design is stable, valid, and ready for hardware deployment in its current form. All major implementation checks passed successfully, including design rule checks, methodology checks, timing analysis, and resource utilization review. Most importantly, the design achieved full timing closure with positive slack across all analyzed paths, confirming reliable operation at the target clock frequency.

As a reduced version of the full v1 core, v1-mini serves as an effective platform for initial hardware validation and baseline performance analysis. Its simplified I/O structure allows the focus to remain on core functionality and implementation quality before introducing additional interfaces.

These results establish a strong foundation for the next stage of development. Future reports on the full RG Sonic32 CPU v1 will expand on this baseline by incorporating the UART interface and documenting any resulting architectural or implementation changes.